

Task: Cleaning k Shoes ($k = 2$)

Coarser instruction $\mathcal{R}^{(3)}$

Finer instruction $\mathcal{R}^{(2)}$

r_1

If the number of dirty shoes is greater than zero, then clean one shoe

$$r_1 : \{N_f > 0\} \mapsto \{N_f \downarrow\}$$

r_{1a}

Pick up the rag

$$r_{1a} : \{\neg H_r\} \mapsto \{H_r\}$$

r_{1b}

If you're holding an unsoaked rag, soak it.

$$r_{1b} : \{H_r, \neg S_r\} \mapsto \{S_r, H_r?\}$$

r_{1c}

With a soaked rag and dirty shoes, clean the shoes.

$$r_{1c} : \{N_f > 0, S_r\} \mapsto \{N_f \downarrow\}$$

r_2

Once the shoes are clean, place the rag on the floor.

$$r_2 : \{N_f = 0, \neg O_r\} \mapsto \{O_r\}$$

